

Haoyu Zhen

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EDUCATION

- University of Massachusetts Amherst** Amherst, US
• *Ph.D. Student* Sept 2024 - 2029 (Expected)
 - Overall GPA: 3.94/4.0
 - Supervisor: Prof. Chuang Gan
 - Research Statement: My research aims to build 3D and 4D world models that enable physical AI to perceive, reason, and act in dynamic environments. By unifying world modeling with reasoning, I seek to develop foundation models that bring robots closer to human-like intelligence in the physical world.
 - Research Impact: GitHub: 2,500+ stars | Google Scholar Citations: 1,500+
- Shanghai Jiao Tong University** Shanghai, China
• *Bachelor of Engineering* Sept 2020 - June 2024
 - Overall GPA: 92.80/100, 4.03/4.30 (Rank 4/88).
 - 4th semester GPA: 95.24/100, 4.20/4.30 (Rank 1/89).
 - Advisor: Cewu Lu and Junchi Yan
- Massachusetts Institute of Technology** Cambridge, US
• *Visiting Student* July 2023 - Dec 2023
 - Supervisor: Prof. Josh B. Tenenbaum

PUBLICATIONS

- Action Images: End-to-End Policy Learning via Multiview Video Generation** arXiv 2026
• Haoyu Zhen, Zixian Gao, Qiao Sun, Yilin Zhao, Yuncong Yang, Yilun Du, Pengsheng Guo, Tsun-Hsuan Wang, Yi-Ling Qiao, Chuang Gan 70+ stars
- SyncWorld: Visual Calibration Enables World Models as Zero-Shot Simulators** arXiv 2026
• Yuncong Yang*, Zhengtao Han*, Furkan Özyurt*, Zeyuan Yang, Han Yang, Junyi Cao, Haoyu Zhen, Yilun Du, Chuang Gan 10+ stars
- 3D-Layout-R1: Structured Reasoning for Language-Instructed Spatial Editing** ECCV 2026
• Haoyu Zhen, Xiaolong Li, Yilin Zhao, Han Zhang, Sifei Liu, Kaichun Mo, Chuang Gan, Subhashree Radhakrishnan
- Fast Spatial Memory with Elastic Test-Time Training** ECCV 2026
• Ziqiao Ma, Xueyang Yu, Haoyu Zhen, Yuncong Yang, Joyce Chai, Chuang Gan 100+ stars
- UniCanvas: A Diffusion-base Unified Model for Text-in-Image Joint Generation** CVPR-W 2026
• Zeyuan Yang, Hao-Wei Chen, Xueyang Yu, Yuncong Yang, Haoyu Zhen, Ziqiao Ma, Maohao Shen, Chuang Gan *Best Paper*
- GHOST: Hierarchical Sub-Goal Policies for Generalizing Robot Manipulation** RSS 2026
• Sriram Krishna, Ben Eisner, Haotian Zhan, Ying Yuan, Haoyu Zhen, Chuang Gan, Shubham Tulsiani, David Held
- Tesseract: Learning 4D Embodied World Models** ICCV 2025
• Haoyu Zhen, Qiao Sun, Hongxin Zhang, Junyan Li, Siyuan Zhou, Yilun Du, Chuang Gan[†] 400+ stars
- 3D-VLA: A 3D Vision-Language-Action Generative World Model** ICML 2024
• Haoyu Zhen, Xiaowen Qiu, Peihao Chen, Jincheng Yang, Xin Yan, Yilun Du, Yining Hong, Chuang Gan[†] 600+ stars
- 3D-LLM: Injecting the 3D World into Large Language Models** NeurIPS 2023 Spotlight
• Yining Hong, Haoyu Zhen, Peihao Chen, Shuhong Zheng, Yilun Du, Zhenfang Chen, Chuang Gan[†] 1,200+ stars
- RapVerse: Coherent Vocals and Whole-Body Motion Generation from Text** ICCV 2025
• Jiaben Chen, Xin Yan, Yihang Chen, Siyuan Cen, Qinwei Ma, Haoyu Zhen, Kaizhi Qian, Lie Lu, Chuang Gan[†]
- Color-NeuS: Reconstructing Neural Implicit Surfaces with Color** 3DV 2024
• Licheng Zhong, Lixin Yang, Kailin Li, Haoyu Zhen, Mei Han, Cewu Lu[†]

- **Relative Entropic Optimal Transport: a (Prior-aware) Matching Perspective to (Unbalanced) Classification** NeurIPS 2023
Liangliang Shi, Haoyu Zhen, Gu Zhang, Junchi Yan[†]
- **CHORD: Category-level in-Hand Object Reconstruction via Shape Deformation** ICCV 2023
Kailin Li^{*}, Lixin Yang^{*}, Haoyu Zhen, Zenan Lin, Xinyu Zhan, Licheng Zhong, Jian Xu, Kejian Wu, Cewu Lu[†]
- **Understanding and Generalizing Contrastive Learning from the Inverse Optimal Transport Perspective** ICML 2023
Liangliang Shi, Gu Zhang, Haoyu Zhen, Jintao Fan, Junchi Yan[†]

EXPERIENCE

- **Meta Platforms, Inc.** Redmond, US
Research intern, Hosts: Zhipeng Cai and Julian Straub
World Models, Video Diffusion Models
May 2026 - Jan 2027
- **NVIDIA Corporation** Santa Clara, US
Research intern, Host: Subhashree Radhakrishnan
Collaborator: Xiaolong Li, Tsung-Yi Lin, Sifei Liu, Zhiding Yu, and Kaichun Mo
3D Vision Language Models, Spatial Reasoning Models
May 2025 - Aug 2025
- **Meshy AI, Taichi Graphics** Beijing, China
Research intern, Host: Tiantian Liu and Yuanming Hu
3D Generation and Diffusion Model
Feb 2024 - Aug 2024
- **Computational Cognitive Science Lab, MIT** Cambridge, US
Visiting scholar, supervised by Josh Tenenbaum
Collaborator: Chuang Gan, Yining Hong, and Zhenfang Chen
Diffusion Model, Large Language Model, World Model, and Embodied AI
Mar 2023 - Dec 2023
- **Machine Vision and Intelligence Group** Shanghai, China
Undergraduate researcher, supervised by Cewu Lu and Lixin Yang
3D Reconstruction and Hand-Object Interaction.
Dec 2021 - July 2024
- **SJTU-ReThinklab** Shanghai, China
Undergraduate researcher, supervised by Junchi Yan.
Sept 2022 - May 2023
≡ Self-supervised Learning (Contrastive Learning) and Optimal Transport

PROJECTS

- **PyBlend: A Python Library for Blender** Aug 2023
Shanghai Jiao Tong University  [GitHub](#).
- **Shanghai Jiao Tong University AI Pilot Class Website** Jan 2022 - Apr 2022
Shanghai Jiao Tong University  [Website](#) or  [GitHub](#)

HONORS AND AWARDS

- 2024. Outstanding Graduate of Shanghai Jiao Tong University.
- 2023. Rong Chang Scholarship (10 winners, 30000¥)
- 2023/2022/2021. Academic Excellence Scholarship of SJTU (top 10%)
- 2021. Huawei Smart Base Scholarship
- 2019. The first prize of the 36th CPHO semi-finals, Beijing Division (Rank: 17, top 1%)

ACADEMIC SERVICE

- **Reviewer:** ICLR, ICRA, TPAMI, ICCV, NeurIPS, ECCV, CVPR, ICML, AAAI, and WACV
- **Teaching Assistant:** CS 230: Computer Systems Principles (Fall 2025), UMass Amherst;
CS 250: Introduction to Computation (Fall 2024), UMass Amherst

PRESENTATIONS AND INVITED TALKS

- 2025/09. Learning 4D Embodied World Models @ Genesis AI
- 2025/07. Building 3D & 4D Foundation Models @ Lumina Embodied AI Community
- 2025/07. Tesseract: Learning 4D Embodied World Models @ RoboPapers
- 2024/04. Building Embodied 3D Foundation Models @ OpenDriveLab, Shanghai AI Lab

SKILLS SUMMARY

- **Proficient:** Python (PyTorch, NumPy, pandas, scikit-learn, etc.), C/C++, Linux, Shell (Bash/Zsh), L^AT_EX, Git, Blender
- **Familiar:** MATLAB, HTML/CSS, etc.